Session 2025-26 Odd Semester

EC-373 Microprocessor Engineering Lab

Problem Statements for writing practice programs in Assembly-Level Language for 8086 Microprocessor

PROBLEM SET-2

(Difficulty level: Mixed)

General Instructions

i. **Students shall solve problem statement nos. 11 to 20 using EMU-8086 emulator.** Please refer to the tutorial-videos shared with you earlier, for getting acclimatized with the emulator-tool.

ii. You shall have to get your program-outputs mandatorily verified by TAs for all the programs, before submitting the lab-file to the faculty. All students in the group must be physically present while getting the outputs verified, and also while submitting the files for evaluation.

iii. The deadline for lab-file submission for this Problem Set-2 will be shared in due course of time.

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List of Problem Statements

Problem No. 11:

Write an assembly-code for 8086-Microprocessor to perform the following steps:

i. Store the following ten 8-bit numbers continuously in memory locations 3000 H to 3009 H using Immediate addressing mode instructions :
F4 H, 02 H, F2 H, 08 H, 04 H, 24 H, 26 H, E8 H, D8 H, 48 H.

ii. Load the following values address values in the registers given below:

BP : 1000 H

SP : 1004 H

SI : 2000 H

DI : 2004 H

iii. Now, store the addition of all ten numbers in successive memory locations (based on the size of the total sum) pointed by the Base Pointer register

iv. Store the difference of first two numbers in the above list, in the memory location pointed by the Stack Pointer Register

v. Store the product of the last two numbers in the above list, in successive memory locations (based on the size of the product) pointed by the Source Index Register.

vi. Store the quotient and remainder of the division of first two numbers in the above list, in successive memory locations pointed by the Destination Index Register.

Problem No. 12:

Write an assembly language program in EMU-8086 emulator, to increment the values of ten numbers stored in the memory locations 5200 H, 5201 H, 5202 H, ……, 5209 H , respectively by values of 1, 2, 3,….., 10. The incremented values of each of these 10 numbers are to be stored back respectively in the same memory locations.

Problem No. 13:

Write an assembly language program-code to find the 1s' complement and 2s' complement of a number stored in memory location 1000 H, and store the results in BX register, and Accumulator register respectively.

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Problem No. 14:

(14-a). Write an assembly language program to two store any two hexadecimal numbers in the following locations: AX and BX. Now, store the result of following two boolean operations (bit-wise operations) in CX register and DX registers respectively:

i. $A \oplus B$

ii. $\overline{A \oplus B}$

(14-b). Write an assembly language program to two store any three 16-bit hexadecimal numbers (A, B, C) in the following locations: AX, BX and CX. Now, store the result of the following boolean operations (bit-wise operations) in DX register:

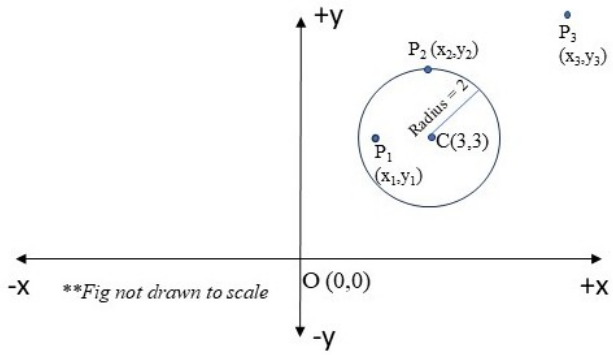
$$F = A.B.C + A.\overline{B} + \overline{A}.C$$

Problem No. 15:

The circle shown in this figure can be modelled by a mathematical equation as :

$(x-3)^2 + (y-3)^2 = 4$

where, center of the circle is at the co-ordinates C(3,3), and the radius (r) of the circle is 2 units.



Write an assembly-level code for an 8086-Microprocessor to detect if a given point with co-ordinates P (x,y) lies on the circle / inside the circle / outside the circle.

Write the program code so as to store different values accordingly in the lower-byte of the BX register in 8086-Microprocessor as follows: Store BL = 11 H (if given point P lies inside the circle) ; Store BL = 00 H (if given point P lies on the circle) ; Store BL = FF H (if given point P lies outside the circle)

(Hint: For a point outside circle: $(x-3)^2 + (y-3)^2 > r^2$; For a point inside circle: $(x-3)^2 + (y-3)^2 < r^2$)

Show your outputs for atleast any three different co-ordinate points as follows :

- P1 (inside the circle)
- P2 (on the circle)
- P3 (outside the circle).

Pre-requisite information for solving Problems 16 & 17:

MOSFET DEVICE MODELING : FUNDAMENTAL CONCEPTS

A MOSFET device is modeled in various regions of operations based on its bias voltages (V_{GS} and V_{DS}) as follows:

When $V_{GS} < V_T$; the MOSFET device is said to be in OFF-state.

When $V_{GS} >= V_T$; the MOSFET device is said to be in ON-state

Also, for a MOSFET in ON-state, the region of operation in the output characteristics is classified as follows, based on the Pinch-off point

$$V_P = V_{GS} - V_T$$

A MOSFET is said to be biased in Linear Region when

$$V_{DS} < V_{GS} - V_T$$

A MOSFET is said to be biased in Saturation Region when

$$V_{DS} >= V_{GS} - V_T$$

Also, based on the semiconductor device physics of operation, we know that the MOSFET transistor device (Metal Oxide Semiconductor Field Effect Transistor) can be modelled with following mathematical equations for its different important regions of operations, namely: SATURATION region and LINEAR region.

The Drain-current of MOSFET device in SATURATION region ($I_{D,sat}$) is given by the following equation :

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$$I_{D,sat} = \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)[V_{GS} - V_{TH}]^2$$

Also, the Drain-current of MOSFET device in LINEAR region ($I_{D,lin}$) is given by the following equation :

$$I_{D,lin} = \mu_n C_{ox}\left(\frac{W}{L}\right)\left[(V_{GS} - V_{TH})V_{DS} - \frac{V_{DS}^2}{2}\right]$$

where, the parameters in the MOSFET-equations can be understood as follows :

μ_n = Mobility of electrons present in the n-channel

C_{ox} = Unit area oxide capacitance

W = Channel Width

L = Channel Length

(W/L) = Aspect Ratio of MOSFET

V_{GS} = Gate-to-Source Voltage applied

V_{DS} = Drain-to-Source Voltage applied

V_T = Threshold Voltage.

Problem No. 16:

Write a 8086 Program code to classify a MOSFET as biased in one of the following three regions of operation, by assuming values for bias voltages (V_{GS} and V_{DS}) and threshold voltage (V_T) :

- OFF state
- Linear Region
- Saturation Region

As the final output, store the value in BL register as follows:

BL = 00 H, if MOSFET is in OFF state

BL = 11 H, if MOSFET is in Linear Region

BL = FF H, if MOSFET is in Saturation Region

In your lab file, you must show outputs *for atleast three different bias conditions for MOSFET, by assuming relevant values for* V_{GS}, V_{DS} *and* V_T.

Problem No. 17:

Assume that you are given a following practical problem to compute the drain-currents of a MOSFET device under both SATURATION region and LINEAR region of operation :

- Take a MOSFET device with an Aspect Ratio (W/L ratio) of 150, and a threshold voltage of 1 V, where both these quantities are inherent device parameters of the MOSFET transistor.
- Assume that the product of the mobility and unit-area oxide capacitance is given by the following constant value:

$$\mu_n \, . \, C_{ox} = 53 \ \mu \text{ F / V.s}$$

You are required to compute the Drain current value of the MOSFET device using a 8086-CPU, under two following regions of operation:

i. LINEAR Region, for given input $V_{GS} = 4$ V ; $V_{DS} = 1$ V

ii. SATURATION Region, for given input $V_{GS} = 4$ V ; $V_{DS} = 4$ V

Write a program-code in Assembly-language in such a way that the input parameters given in the above problem statement are stored in the following RAM memory locations for the purpose of computation by 8086-CPU. Also, ensure that the two output values ($I_{D,lin}$ & $I_{D,sat}$) are also stored back in the respective memory locations as given below.

(Hint: Use the value given in the units as it is, so as to obtain the two values of drain current in terms of micro-amperes, when the program-code is being executed in the EMU-8086 emulator software).

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RAM-Location	1000 H	1001 H	1002 H	1003 H, 1004 H	1005 H, 1006 H	1007 H & 1008 H	1009 H & 100A H
Stored Value	**Aspect Ratio**	**μ_n . C_{ox} (in unit µF/V.s)**	**Thres-hold Voltage**	**Gate to Source Voltages**	**Drain to Source Voltages**	**Drain current in Lin-Reg ($I_{D,lin}$)**	**Drain current in Sat-Reg ($I_{D,sat}$)**

Problem No. 18:

Write an assembly language program code for 8086-Microprocessor, to compute the value of the following exponential function:

$$\mathbf{f(x) = e^x\ ,\ at\ the\ point\ \ x = 3,}$$

where "e" is the transcendental number, more famously called as the "Euler's number" in mathematics. Use the following Mac-Laurin's series expansion for the exponential function, which is obtained in calculus as the special case of Taylor's series expansion of e^x where successive higher order derivatives are computed at the point '0' :

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots + \frac{x^n}{n!} + \ldots\ldots\infty$$

Use a conventional calculator to compute the value of e^3 corresponding to LHS in the above equation.

Now, write an assembly program to obtain the value of RHS of above equation at the point x=3, using the EMU-8086 emulator, with values computed in an

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iterative program loop atleast upto 5th power in the Mac-Laurin's series expansion.

Compare the results with the values obtained from the calculator and the EMU-8086 emulator.

Problem No. 19 :

Write an assembly-language program-code to solve the following solid state device problem, and implement the same in 8086 Microprocessor Hardware kit present in your lab work-bench.

You are asked to calculate the forward-bias current of a Silicon PN junction diode (in μA), under given experimental conditions, modelled through a Mac-laurin's series expansion using an 8086-Microprocessor. Compare the approximate-value of diode-current (in μA) obtained from the 8086-Microprocessor with that of the more accurate-value (in μA) obtained from a conventional calculator using the exponential function in Shockley's equation.

The experimental conditions of the Silicon-diode under forward-bias are given as follows:

Forward bias applied voltage, $V_D = 260\ mV$; *Diode ideality factor,* $\eta = 2$;

Thermal voltage, $V_T = 26\ mV$; *Reverse saturation current,* $I_O = 1\ \mu A$

Hints: Take Shockley's equation to model the diode's behaviour in its approximate form as follows: Diode-current, $I_D \sim I_O . exp(V_D / \eta V_T)$

Also, from your knowledge of Calculus, use the Mac-laurin's series expansion with an accuracy till 3rd power as follows, to write the program-code in 8086 Microprocessor:

$$e^x = 1 + x + (x^2 / 2!) + (x^3 / 3!)$$

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Problem No. 20:

Computing "Approximate Small-signal performance of a BJT" using "8086-Microprocessor"

In Mathematics, the transcendental number "e" has got much significance, and is a very interesting quantity to explore..!!

We know that this transcendental number takes the value

$$e = 2.718281\ldots\ldots$$

We also know, from high-school calculus that, for the function $f(x) = e^x$, we can write a Mac-Laurin's Series expansion as below (which is a special case of Taylor's series expansion) :

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots + \frac{x^n}{n!} + \ldots\ldots\infty$$

……EQN-1

Now, let us re-collect some interesting applications of the above mathematical concept with regards to modelling of Bipolar Junction Transistors in performing AC analysis. We know that, for AC analysis, "approximations" are used to derive "Small-Signal equivalent models" for the transistor-device..!!

We know that the effective Base Current (i_B) of a Bipolar Junction Transistor can be modeled with following equation for the Base-emitter diode present in the Bipolar Junction Transistor (biased in common-emitter configuration), where the effective base current is represented as a combination of a DC bias term and an AC small-signal term as follows :

$$\mathbf{i_B = I_{BQ}.\ exp\ (v_{be} / \eta V_T)} \quad \ldots\ldots\ldots\ldots\ldots\ldots\text{EQN-2}$$

where,

i_B = Effective Base-current of BJT (DC + AC value)

v_{be} = Small-signal base-emitter voltage (purely AC value)

I_{BQ} = DC Base-Current of BJT at the given bias Q-point

η = Ideality factor, taken as 2 for Silicon BJT

V_T = Thermal voltage given by kT/q, taken as 26 mV at

standard room temperature

Hence, by writing Mac-Laurin's series (EQN-1) expansion for the exponential term present in EQN-2, we shall obtain the following:

$$i_B = I_{BQ}\left[1 + \frac{v_{be}}{\eta V_T} + \frac{1}{2!}\left(\frac{v_{be}}{\eta V_T}\right)^2 + \frac{1}{3!}\left(\frac{v_{be}}{\eta V_T}\right)^3 + \ldots\ldots + \frac{1}{n!}\left(\frac{v_{be}}{\eta V_T}\right)^n + \ldots\ldots\right]$$

……..EQN-3

Considering the above theoretical concepts, now take the following practical BJT-problem, and solve it using the 8086-Microprocessor :

Assume that a Silicon BJT is biased in common-emitter mode with a DC bias base-current value of 40 μA, in standard room temperature. Assume that an instantaneous AC small-signal voltage of 104 mV is applied to the base-emitter junction (in CE-mode). For these given conditions, calculate the resultant effective value of base current in the BJT (i_B), upon applying this small signal AC voltage at the input base-side. Use the exponential equation (EQN-2) to plug-in the given values, and hence calculate the numerical value of i_B, rounded to the nearest integer. For this calculation, use a conventional calculator to compute the value of the exponential term, and hence the final rounded value of i_B.

Now, use the 8086-Microprocessor to cross-verify the above calculated value of i_B. To do this, write an assembly language program using suitable iterative-loops for computing the value corresponding to RHS in EQN-3, with an approximation of atleast till the 7th power term in the Mac-Laurin's series expansion for the BJT-equation. Load the four numerical values given in the problem-statement, namely: I_{BQ}, V_T, η, v_{be} onto four different

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memory locations in the RAM. Finally, store the computed value of Base-current, i_B, onto a separate RAM location.

Compare & conclude your results in the lab-file with the following two outputs :

i. **The more accurate value of i_B computed using the exponential function directly using a conventional calculator**

ii. **The approximate value of i_B computed using the Assembly-language program by the 8086-Microprocessor**

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